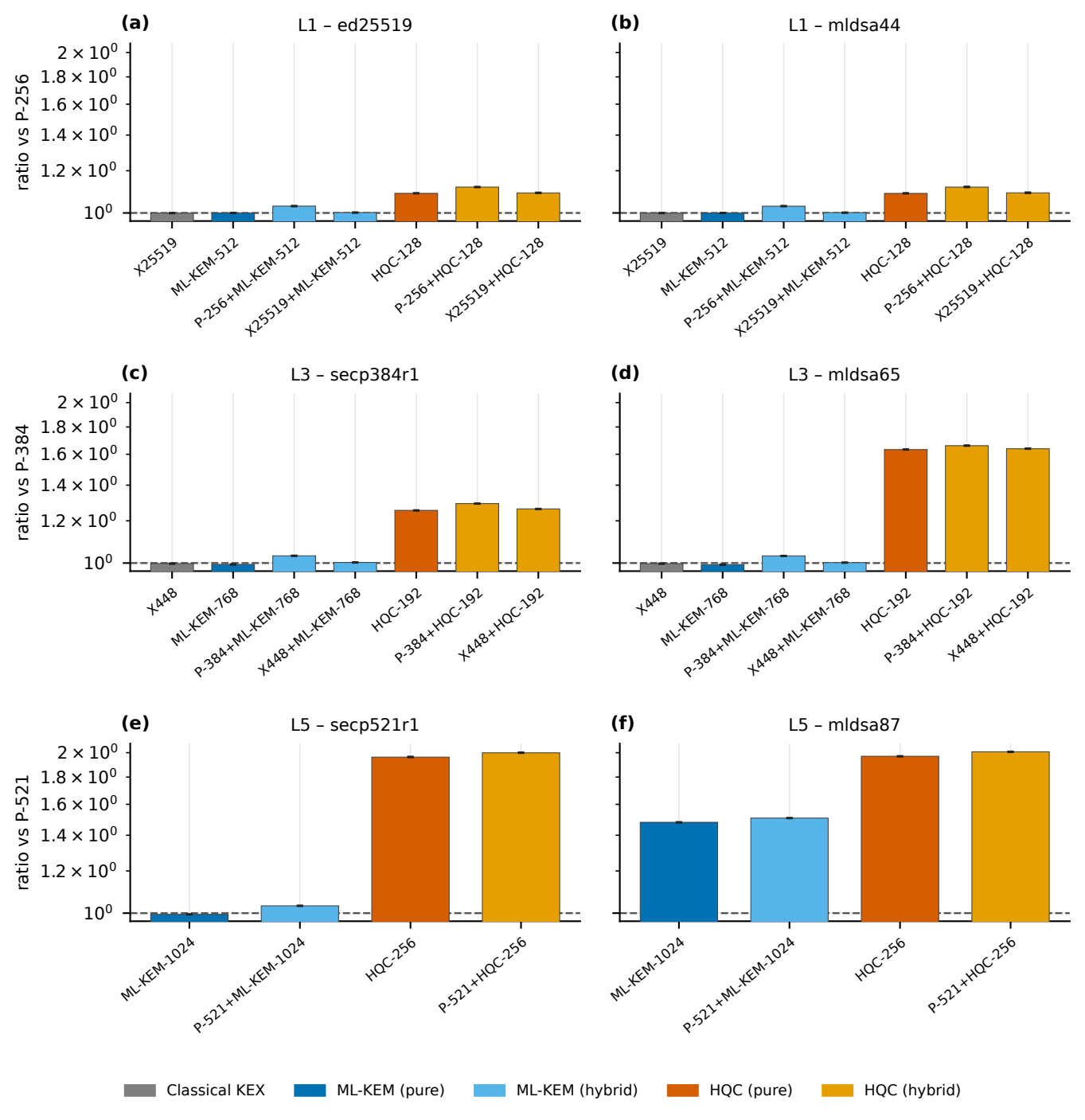
(a)
L1 – ed25519
(b)
L1 – mldsa44
(c)
L3 – secp384r1
(d)
L3 – mldsa65
(e)
L5 – secp521r1
(f)
L5 – mldsa87
ratio vs P-256
ratio vs P-384
ratio vs P-521
2 × 10^0
1.8 × 10^0
1.6 × 10^0
1.4 × 10^0
1.2 × 10^0
10^0
X25519
ML-KEM-512
P-256+ML-KEM-512
X25519+ML-KEM-512
HQC-128
P-256+HQC-128
X25519+HQC-128
X448
ML-KEM-768
P-384+ML-KEM-768
X448+ML-KEM-768
HQC-192
P-384+HQC-192
X448+HQC-192
ML-KEM-1024
P-521+ML-KEM-1024
HQC-256
P-521+HQC-256
Classical KEX
ML-KEM (pure)
ML-KEM (hybrid)
HQC (pure)
HQC (hybrid)